

Lemma 2.3. For a binary (near) equireplicate $r \times c$ row-column design on v symbols,

$$(a) \lambda_{rc} = e^- + e^+ - \frac{e^- e^+}{e} = e + \frac{(e^+ - e)(e - e^-)}{e},$$

$$(b) \lambda_{rc}^- = e^- \text{ and } \lambda_{rc}^+ = e^+,$$

$$(c) \lambda_{rr} = \frac{c(\lambda_{rc} - 1)}{r - 1},$$

$$(d) \lambda_{cc} = \frac{r(\lambda_{rc} - 1)}{c - 1}.$$

Remark 2.4. Lemma 2.3 (a) implies that $\lambda_{rc} \geq e$, and that the equality holds if and only if e is an integer, i.e. the row-column design is equireplicate.

Proof of Lemma 2.3. For an equireplicate design all claims hold due to Lemma 2.2, so from now on we assume that it is near equireplicate, that is, $e^- < e < e^+$. Noting that $v = v_- + v_+$, a double counting argument similar to the proof of Lemma 2.2 together with Lemma 2.1 gives that the number of common symbols summed over all pairs of a row and a column is

$$rc\lambda_{rc} = v_-(e^-)^2 + v_+(e^+)^2 = v(e^+ - e)(e^-)^2 + v(e - e^-)(e^+)^2.$$

Dividing by v and using $e = rc/v$, we get

$$e\lambda_{rc} = (e^+ - e)(e^-)^2 + (e - e^-)(e^+)^2 = e((e^+)^2 - (e^-)^2) + e^-e^+(e^- - e^+). \quad (1)$$

Note that $e^+ - e^- = 1$, so $(e^+)^2 - (e^-)^2 = e^+ + e^-$, thus

$$e\lambda_{rc} = e(e^+ + e^-) - e^-e^+,$$

from which (a) follows. In order to prove (b) it suffices to show that $e^- < \lambda_{rc} < e^+$, which follows from (a):

$$\begin{aligned} e^- &< e < e + \frac{(e^+ - e)(e - e^-)}{e} = \lambda_{rc}, \\ \lambda_{rc} &= e + \frac{(e^+ - e)(e - e^-)}{e} = e + (e^+ - e) \cdot \frac{e - e^-}{e} < e + (e^+ - e) = e^+. \end{aligned}$$

Again using double counting and Lemma 2.1, the number of common symbols summed over all pairs of rows is

$$\binom{r}{2}\lambda_{rr} = v_- \binom{e^-}{2} + v_+ \binom{e^+}{2} = v(e^+ - e) \binom{e^-}{2} + v(e - e^-) \binom{e^+}{2}.$$

Dividing by $v/2$, we get

$$\begin{aligned} \frac{r(r-1)\lambda_{rr}}{v} &= (e^+ - e)e^-(e^- - 1) + (e - e^-)e^+(e^+ - 1) \\ &= e((e^+)^2 - (e^-)^2) + e^+e^-(e^- - e^+) + e(e^- - e^+) \\ &= e\lambda_{rc} - e, \end{aligned}$$

where the last equality follows from (1). This implies (c), and a similar count gives (d). $\square$

Having done this ground work, we now formally define our main object of study.

Definition 2.5. An $(r \times c, v)$ -near triple array is a binary (near) equireplicate $r \times c$ row-column design on v symbols in which

1. any row and column have either λ_{rc}^- or λ_{rc}^+ common symbols,
2. any two rows have either λ_{rr}^- or λ_{rr}^+ common symbols,
3. any two columns have either λ_{cc}^- or λ_{cc}^+ common symbols.

Note that the definition allows a near triple array to be equireplicate. To avoid having to calculate the quantities involved in the definition explicitly, it will often be convenient to instead use the following alternative characterization of near triple arrays.